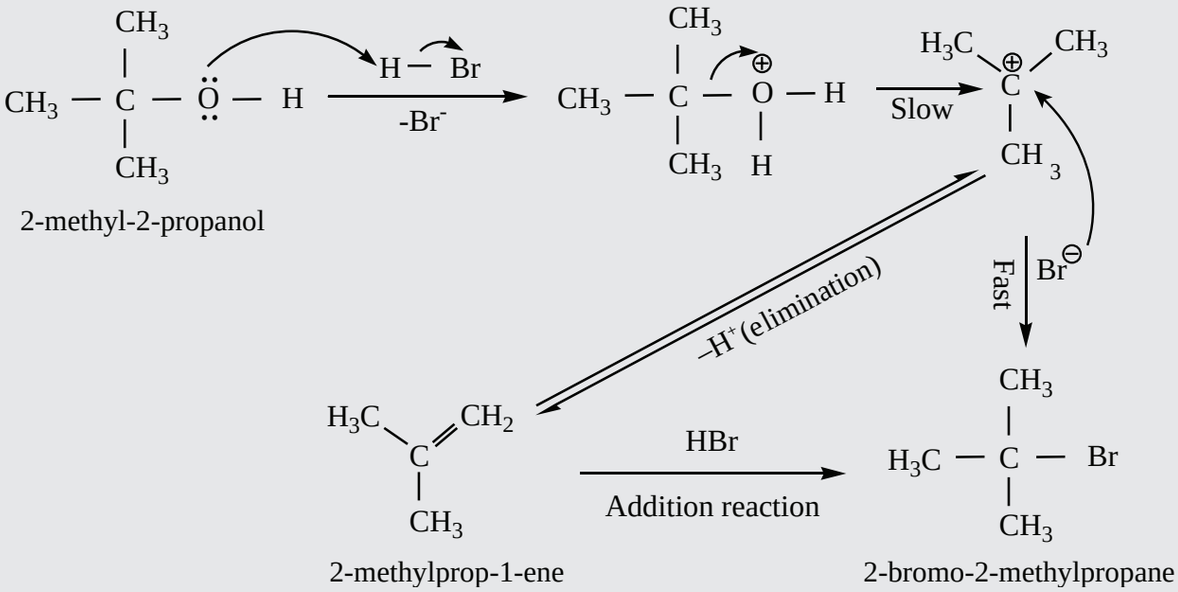
Alkyl halide formation of tertiary alcohols follow S_N1 mechanism.

Example



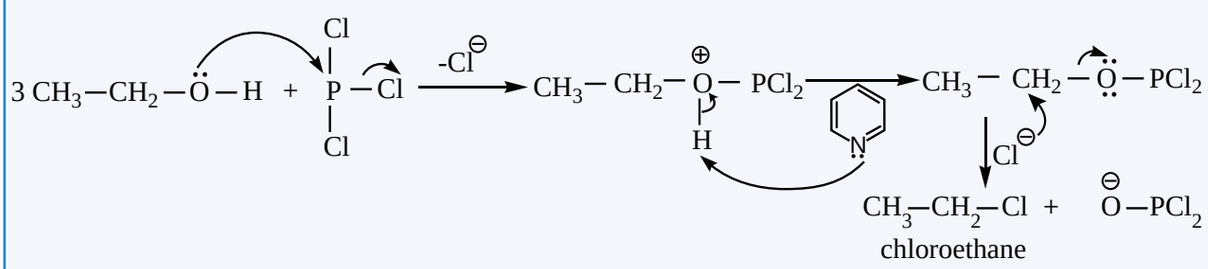
Here, the carbocation formed can undergo elimination to give an alkene. However the alkene can again undergo addition reaction with HBr to give the substituted product.

Conversion of alcohol into alkyl halides: Other methods

Alcohols can also be converted into an alkyl halides using PCl_3, PBr_3.

$$\underset{\text{ethanol}}{CH_3CH_2OH} \xrightarrow[\text{pyridine}]{PCl_3} \underset{\text{chloroethane}}{CH_3CH_2Cl}$$

Mechanism : S_N2 reaction on phosphorous tri chloride



The conversion of an alcohol to alkyl halide can also be effected using thionyl chloride

$$\underset{\text{Methanol}}{CH_3-OH} + \underset{\text{Thionyl chloride}}{SOCl_2} \xrightarrow{\text{Pyridine}} \underset{\text{chloromethane}}{CH_3-Cl} + HCl + SO_2$$

This reaction also follows the S_Ni mechanism in the presence of pyridine.

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